Quantum Fault Tolerance Meets Toom's Contours

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Change letters: the d in the model section refers to a size that is comparable to the code distance where here d refers to the dimension of the Toric complex

1 The (d, d')-Dimensional Toric Code and the SWEEP Rule

Definition 1.1 (d-Dimensional Toric Complex)**.** For integers d and n, denote the group resulting from the d-directed power of the cyclic group of order n by $\mathcal{G}_n^d = \mathbb{Z}_n^{\times d}$, and denote the standard generating set of $\mathcal{G}_n^d$ by $S = \{1_i : i \in [d]\}$. We define the *d-dimensional toric complex* to be the hypergraph obtained from the Cayley graph $\mathrm{Cay}(\mathcal{G}_n^d, S)$ by taking its vertices as the 0-faces and for any $i \in [d-1]$ we define the i-faces as follow: First, for a vertex $v \in \mathcal{G}_n^d$ and for a subset of generators $S' \subset S$ of size $|S'| = i$, the i-face rooted at v and spanned by S', denoted by $\theta_{v,S'}$, is:

$$\theta_{v,S'} := \bigcup_{I \subset S'} \left\{ \left(\prod_{g \in I} g \right) v \right\}$$

The i-faces of the complex are the collection of all the possible rooted spanned i-faces induced by $\mathrm{Cay}(\mathcal{G}_n^d)$. We will denote them by $\mathcal{F}_i$.

Additionally, we name n and d, the *side length* and the *dimension* of the complex.

Definition 1.2 (Positive Edges and Cubes)**.** For $u, v \in \mathcal{G}_n^d$, and $S'_1, S'_2 \subset [d]$, with sizes $|S'_1| = |S'_2| = i$, we say that the i-faces θ_{v,S'_1} and θ_{u,S'_2} are adjacent if the intersection $\theta_{v,S'_1} \cap \theta_{u,S'_2}$ is an $i-1$-face. In addition, if θ_1 and θ_2 are i and $i-1$-faces such that $\theta_2 \subset \theta_1$, then we say that θ_2 is a *positive edge relative to face* θ_1 if they are rooted at the same vertex. Furthermore, we say that θ_1 is a *positive cube* of θ_2. When the i-face is clear from the context, we will abbreviate and say that θ_2 is *positive*.

Example 1.3. For example, let $v, u \in \mathcal{G}_n^d$ and $g_1 \neq g_2 \in S$. Consider the 2-face $\theta = \{v, g_1 v, g_2 g_1 v, g_2 v\}$. Then, for $x, y \in \mathcal{G}_n^d$, we say that an edge (1-face) $\{x, y\}$ is *positive relative to the face* θ if $x = v$ and y is either $g_1 v$ or $g_2 v$.

We denote the complex by $G = (V, E, F)$, where V, E, and F are the 0, 1, and 2 faces.

Add a paragraph that explains that we choose not to use a chain-complex since we earn nothing from it i.e don't care about topology properties. Yet mention that chain-complex formulation proved itself as an effective tool to compute the properties of the codes we use.

Definition 1.4 (Assignment on The (d', d)-Toric Complex)**.** Let $\mathcal{T}$ be a d-dimensional complex. Each of its i-face sets naturally induces a vector space $\mathbb{F}_2^{|\mathcal{F}_i|}$. We call a binary vector $z \in \mathbb{F}_2^{|\mathcal{F}_i|}$ an assignment on the i-faces of $\mathcal{T}$. For a vertex v and a subset of generators $S' \subset S$ of size $|S'| = i$, we use the notation $\theta_{\mathbf{v},\mathbf{S'}}$ to represent the unit vector in $\mathbb{F}_2^{|\mathcal{F}_i|}$ that has 0 in every coordinate except the one associated with the i-face $\theta_{v,S'}$. Then

$$\left\{\theta_{\mathbf{v},\mathbf{S'}} : v \in \mathcal{F}_0, S' \subset S \text{ s.t } |S^{'}| = i\right\}$$

is a base to $\mathbb{F}_2^{|\mathcal{F}_i|}$ Not exactly base, they are linear dependent. Let z be an assignment on the i-faces. Let $v_1, \dots, v_k \in \mathcal{F}_0$ and $S_1, \dots, S_k \subset S^{\times k}$ be the generator subsets such that $\theta_{v_1,S_1'}, \theta_{v_2,S_2'}, \dots, \theta_{v_k,S_k'}$ are the collection of i-faces that support z, namely:

$$z = \sum_{i=1}^{k} \theta_{\mathbf{v_i},\mathbf{S_i'}}$$

We refer to $\theta_{v_1,S_1'}, \theta_{v_2,S_2'}, \dots, \theta_{v_k,S_k'}$ as the collection of faces that form z, or briefly, the faces form z. For $i > 0$, we define the linear map $\partial : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i-1}|}$ such that

$$\partial\theta_{\mathbf{v},\mathbf{S'}} := \sum_{g\in S'} \theta_{\mathbf{v},\mathbf{S'/g}} + \sum_{g\in S'} \theta_{\mathbf{gv},\mathbf{S'/g}}$$

For $i < d$, we define the linear map $\partial^* : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i+1}|}$ such that

$$\partial^*\theta_{\mathbf{v},\mathbf{S'}} := \sum_{g\in S/S'} \theta_{\mathbf{v},\mathbf{S'\cup g}} + \sum_{g\in S'} \theta_{\mathbf{g^{-1}v},\mathbf{S'\cup g}}$$

We define the *restriction of z to a vertex v* to be the linear map $|_v : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_i|}$ such that:

$$\theta_{\mathbf{u},\mathbf{S'}}|_v := \begin{cases} \theta_{\mathbf{u},\mathbf{S'}} & v \in \theta_{u,S'} \\ 0 & \text{else} \end{cases}$$

We define the *front of z to a vertex v* to be the linear map $|_{v+} : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_i|}$ such that:

$$\theta_{\mathbf{u},\mathbf{S'}}|_{v+} := \begin{cases} \theta_{\mathbf{u},\mathbf{S'}} & u = v \\ 0 & \text{else} \end{cases}$$

We say that an assignment z is *connected* if the graph that is induced by taking the faces that support z as vertices and defining the edges to be the adjacency relation between them is connected. We define the connected components of z to be its subsets such that each of them is a connected assignment. For a j-face f, we say that $f \in z$ if z has a support on $\theta_{\mathbf{v},\mathbf{S'}}$ such that $f \in \theta_{v,S'}$. Consider an assignment $z \in \mathbb{F}_2^{\mathcal{F}_i}$ and a vertex $v \in \mathcal{G}_n^d$. We say that z is a *rooted assignment* at vertex v if there are subsets $S'_1, \ldots, S'_k \subset S^{\times k}$ such that each of S'_j is at size i and z is decomposed as:

$$z = \sum_{j}^{k} \theta_{\mathbf{v},\mathbf{S'_j}}$$

Put differently, z is the result of taking a collection of i-faces that are rooted at v.

Add a proof for $\partial\theta_{\mathbf{v},\mathbf{S'}} \cdot \partial^*\theta_{\mathbf{u},\mathbf{S''}} = 0 \Rightarrow$ gives a CSS code.

Need to decide on what should I elaborate, for example should I provide a proof for the distance and the rate of the code? [Den+02]

Definition 1.5 ((d, d') Toric Code)**.** Let d, d' be integers such that $1 < d' < d$. For any integer n, we construct the nth instance of the quantum code family (d, d') *Toric Code* as follows: Let $G = (\mathcal{F}_0, \ldots, \mathcal{F}_d)$ be the d-dimensional toric complex obtained from $\mathcal{G}_n^d$ as defined in Definition 1.1. We associate the (q)bits with the $\mathcal{F}_{d'}$, the X-checks with the $\mathcal{F}_{d'-1}$, and the Z-checks with the $\mathcal{F}_{d'+1}$. An X-check, c_x, is satisfied if the parity of (q)bits set on the d'-face containing c_x is even. Similarly, a Z-check, c_z, is satisfied if the parity, in the Hadamard basis, of the (q)bits set on the d'-faces contained in c_z is even.

Definition 1.6 (Sweep Rule)**.** Let z be an assignment over the d'-faces, The *sweep rule decoding procedure* at vertex $v \in \mathcal{G}_n^d$, defining as follow:

1. For the $\mathcal{C}_X$ code. Look for a rooted assignment $\tilde{z}$ at v such that $(\partial\tilde{z})|_{v+} = \partial(z)|_{v+}$. Namely, the X-checks rooted at v get the same syndromes for z and $\tilde{z}$. If there is one, then flip $\tilde{z}$.

2. Apply the Hadamard gate on any d' face containing v.

3. For the $\mathcal{C}_Z$ code. Look for a rooted assignment $\tilde{z}$ at v such that $(\partial^*\tilde{z})|_{v+} = \partial^*(z)|_{v+}$. Namely, the Z-checks rooted at v get the same syndromes for z and $\tilde{z}$. If there is one, then flip $\tilde{z}$.

4. Apply the Hadamard gate again on any d' face containing v to reverse the second stage.

Stages 1 and 3 involve measurement of stabilizers.

Definition 1.7 (Decoding Gadget)**.** Consider a quantum circuit C, for a quantum gate u in C, with fun-in Δ, we say that u is a *decoding gadget* if it has the following form:

1. The wires of u wires can be decomposed into two subsets A and B such that the wires in A are reset wires, and there exists an encoded register R in C such that the wires in B are all set in R.
2. For a set of faults $\mathcal{E}$, there is a Pauli gate u' that acts only on B such that the action of u in $C[\mathcal{E}]$ equals u'. Put differently, the quantum circuit obtained from $C[\mathcal{E}]$ by replacing u with u' equals $C[\mathcal{E}]$.

Definition 1.8 (Unitary Decoding Representation)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit, and let $\mathcal{U} = \{u'_i\}_{i=1} \subset C$ be a subset of the gates in C that are decoding gadgets. For a set of faults $\mathcal{E}$, we define the *unitary decoding representation* of $C[\mathcal{E}]$ to be the quantum circuit obtained by replacing each of the gates in $\mathcal{U}$ with their corresponding Pauli gate according to $\mathcal{E}$.

Claim 1.9. The local gates performing Toom's rule are decoding gadgets.

Proof. prove that. □

Definition 1.10 (Base graph G_o)**.** Let C be a quantum circuit, at depth d, with registers $R_1, \dots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, .., \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \dots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:
$$V_o := \big\{(v, t) : t \in [d], v \in \bigcup_i \mathcal{G}_i\big\}$$
2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \dots, \mathcal{G}_k$ with the addition that any two vertices, in preceding times t and $t+1$, that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:
$$\begin{aligned} E_o := \big\{\{(v,t),(u,t')\} : \text{if} & \\ & \text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ & \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1t\big\} \end{aligned}$$

2 Infinite Torics

Definition 2.1 (Registers Coordinates System)**.** Let $C = \{u_i\}_{i=1}$ be a quantum circuit, with width w and registers $\mathcal{R} = R_1, \dots, R_k$, each of length at most n. For register R_i, we define the register identifier $\mathbf{1}_{R_i} = \mathbf{1}_i$. Then, we define the map $\phi_{\mathcal{R}} \colon \mathbb{Z}_w \to \mathbb{Z}_R \times \mathbb{Z}_n$ to return, for a space-location coordinate $x \in \mathbb{Z}_w$, the tuple consisting of the identifier of the register containing x, and its relative coordinate within the register. Namely,

$$\psi_{\mathcal{R}}(x) = (\mathbf{1}_{R_j}, x')$$

Where R_j is the register containing x, and x' is the index of the qubit, set at the x location, in R_j. The presentation of C in the *registers coordinate system* is the encoding by a set of gates $\{u'_j\}_{j=1}$, which is the result of replacing each space-location of the original gates with the applications of $\psi_{\mathcal{R}}$ on them. Namely, for each $u_i = (g_i, l_i) \in C$, we map its space-location coordinate $l_{i_2} = (l_{i_2}^{(1)}, \dots, l_{i_2}^{(\Delta)})$ to:

$$l_{i_2} \mapsto (\psi_{\mathcal{R}}(l_{i_2}^{(1)}), .., \psi_{\mathcal{R}}(l_{i_2}^{(\Delta)}))$$

For brevity, we name the first and second coordinates in the result of ψ_R the *register coordinate* and the *inner coordinate.*

Definition 2.2 (Shifting in Registers System)**.** Consider the register coordinate system induced by a quantum circuit of width w, with R registers, each of length at most n. For an integer $r \in \mathbb{Z}$, we define the r-shifting function $\mathrm{Sh}_r\colon (\mathbb{Z}_R \times \mathbb{Z}_n)^{\times\Delta} \to (\mathbb{Z}_R \times \mathbb{Z})^{\times\Delta}$ to act on space-location coordinates as follows:

$$\mathrm{Sh}_r(l_1, .., l_k) = (\mathrm{Sh}_r(l_1), .., \mathrm{Sh}_r(l_k))$$
$$\text{Where } \mathrm{Sh}_r(l_i) = \begin{cases} l_i + r & \text{if } l_i \text{ is an inner coordinate} \\ l_i & \text{else} \end{cases}$$

Definition 2.3. Let n be an integer, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with registers $\mathcal{R} = R_1, \dots, R_k$ such that each register R_i is a toric encoding with a side length n. In addition, all the decoding gadgets of C are sweep rule gadgets; let us denote them by $\mathcal{U}$. We define the ∞-*extension of* C, denoted by C^∞, as follows:

1. For a register $R_i \in \mathcal{R}$, denote by d_i, d'_i the dimension of the toric complex that induces the code, and the dimension of the faces that are associated with the bits. Each register R_i in $\mathcal{R}$ is mapped to a register R'_i, which is a (d_i, d'_i)-dimensional, infinite side length, toric encoding. Since the mapping between registers is one-to-one, we use the identifier $\mathbf{1}_{R_i}$ to identify also R'_i.

2. Each gate $u_i \in C/\mathcal{U}$ is mapped to:

$$u_i = (g_i, l_i) \mapsto \bigcup_{j=-\infty}^{\infty} \{(g_i, (\mathrm{Sh}_{jn}(l_i))\}$$

3. Each gate $u \in \mathcal{U}$ is mapped to the collection of sweep rule gadgets $\{v_j\}_{j=-\infty}^{\infty}$ such that each of them has the same time coordinate as u, is in the register that corresponds to the image of u's register, and is rooted at $\mathrm{root}(v) + jn$.

For a set of faults $\mathcal{E} = \{f_i\}_{i=1}$, we denote by $C^\infty[\mathcal{E}^\infty]$ the ∞-*extension of* $C[\mathcal{E}]$.

Claim 2.4. Let $u = (g, l)$ and $v_0, v_{\pm 1}$ plantation of u at locations $\text{root}(u)$ and $\text{root(u)} \pm n$. Then $v_0 v_{\pm 1}|_{[w]} = u$.

Proof. Prove it $\square$

Definition 2.5. For a quantum circuit C, of the form defined in Definition 2.3, and a set of faults $\mathcal{E}$, denote:

$$\rho := C[\mathcal{E}] \, |0\rangle\langle 0| \, C[\mathcal{E}]^\dagger$$
$$\rho_\infty := C^\infty[\mathcal{E}^\infty] \, |0\rangle\langle 0| \, (C^\infty[\mathcal{E}^\infty])^\dagger$$

We say that a quantum circuit C is ∞*-compatible* if, for any set of faults $\mathcal{E} = \{f_i\}_{i=1}$, it holds that:

$$\rho = \mathbf{Tr}_{/[w]} \left(\rho_\infty\right)$$

Claim 2.6. Let C be quantum circuit of the form defined in Definition 2.3. Then C is ∞-compatible.

Proof. Its enough to prove the claim for circuits in their unitary decoding representation. We prove it by induction on the number of the gates.

1. Base. For an empty circuit, ρ_∞ is just the infinite string of zeros; thus, the restriction of it to $[w]$ locations, namely $\mathbf{Tr}_{/[w]}(\rho_\infty)$, yields a single matrix element corresponds to the w-length string of zeros, which is exactly ρ.

2. Step. Assume the correction of the claim for any circuit, with less than τ steps, and consider a circuit C with τ steps. Denote by C' the circuit obtained form C by erasing the last gate in C, observes that C' is a circuit with $\tau - 1$ steps. Denote by ρ' and ρ'_∞ the: Enter it. By the induction assumption we have:

   $$\rho' = \mathbf{Tr}_{/[w]} \left(\rho'_\infty\right)$$

   Denote the last gate in C by u, and by $I = \{v_j\}_{-\infty}^{\infty}$ the collection of gates in C^∞ to which u is mapped. Let us split into cases, depending on whether u is or is not a decoding gate.

   (a) Suppose that u is not a decoding gate. Let g and l be the gate element and the location of u. By the definition of ∞-extension, I equals:

   $$I = \bigcup_{j=-\infty}^{\infty} \{(g, (\text{Sh}_{jn}(l))\}$$

   So there is only one gate in I, corresponding to $j = 0$, that has support on qubits with spatial location in the range $[w]$. Denote that gate by v_0.

$$\begin{aligned}
\mathbf{Tr}_{/[w]}\left(\rho_\infty\right) &= \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty}^{\infty} v_j \rho'_\infty \prod_{j=-\infty}^{\infty} v_j^\dagger\right) \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty, j\neq 0}^{\infty} v_j \rho'_\infty \prod_{j=-\infty, j\neq 0}^{\infty} v_j^\dagger\right) v_0^\dagger \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\prod_{j=-\infty, j\neq 0}^{\infty} v_j^\dagger \prod_{j=-\infty, j\neq 0}^{\infty} v_j \rho'_\infty\right) v_0^\dagger \\
&= v_0 \mathbf{Tr}_{/[w]}\left(\rho'_\infty\right) v_0^\dagger \\
&= v_0 \rho' v_0^\dagger \\
&= \rho
\end{aligned}$$

(b) In the case that u is a decoding gate, observe that there are at most two gates in I that have support on the qubits with space-location in $[w]$. If there is exactly one, then repeating the non-decoding gate case while setting this gate as v_0 gives the desired result. Else, in the case where two gates act non-trivially on qubits at $[w]$, denote those gates by v_0, v_1. Since we assumed that C is given in its unitary decoding representation, it holds that v_0, v_1 are Paulis, in particular a product operator. Denote their product decompositions by $v_0 = \prod_k v_0^{(k)}$ and $v_1 = \prod_k v_1^{(k)}$. Thus, by repeating the non-decoding case, but extracting from the infinite product only the terms in v_0 and v_1 that act non-trivially at $[w]$, we get:

$$\mathbf{Tr}_{/[w]}\left(\rho_\infty\right) = v_0 v_1|_{[w]} \rho' \left(v_0 v_1|_{[w]}\right)^\dagger$$

Moreover, since v_0 and v_1 are a plantation of g in locations $\mathrm{root}(u)$ and $\mathrm{root}(u) \pm n$, then by Claim 2.4, we have that $u = v_0 v_1|_{[w]}$; therefore:

$$\begin{aligned}
\mathbf{Tr}_{/[w]}\left(\rho_\infty\right) &= v_0 v_1|_{[w]} \rho' \left(v_0 v_1|_{[w]}\right)^\dagger \\
&= u \rho' u^\dagger \\
&= \rho
\end{aligned}$$

□

add that the computation graph is lifted naturally to the ∞-extension, since the image of a mapping are overlapped gates, we can execute them simultaneously.